



Figure 7: Organisations developing PyTorch

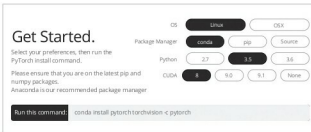


Figure 8: Installation instructions on different parameters

PyTorch: Deep learning in Python

PyTorch (<http://pytorch.org/>) is a free and open source framework for DL using Python programming. Dynamic models and neural networks can be programmed using PyTorch. It supports CPU as well as GPU based computing for DL algorithms.

PyTorch provides the following features for high performance DL:

- Tensor processing and computation with GPU support
- Deep neural networks
- Reusability of Python scientific packages on demand like SciPy, NumPy, scikit-learn and Cython
- Minimal overhead and complexities
- Fast processing regardless of the size of the neural network in computation
- Integration with acceleration and speed-up libraries including Intel MKL and NVIDIA to escalate the speed and minimise delay

The list of organisations, universities and companies working on developing PyTorch is shown in Figure 7.

The installation process of PyTorch varies depending upon the operating system and the package manager, as specified in the official portal of PyTorch. The developers can install PyTorch by selecting the OS, package manager, Python version and CUDA support, as shown in Figure 8.

Example of the classifier using deep learning in PyTorch

The features of deep learning are presented with a

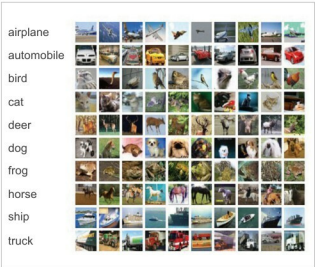


Figure 9: The benchmark data set of CIFAR-10



Figure 10: Testing the data set for classification and prediction

classification, using a benchmark data set CIFAR-10 as shown in Figure 9 (<https://www.cs.toronto.edu/~kriz/cifar.html>). This data set is used for training the model of deep learning. In this data set, multiple similar images are labelled. Using PyTorch, the image that does not exist in the data set can be predicted under a specific class and label category.

The output will be fetched as 'plane horse cat bird' because of the feature extraction and deep learning, based on the properties of these objects extracted from the training data set. Despite the fact that none of the images in Figure 10 exist in the training data, the deep learning algorithm is able to identify similar patterns and assign the class or label to the appropriate category.

Scope for research and development

Deep learning libraries can be used for R&D in assorted applications including criminal face detection, flying objects analysis, intrusion detection at international borders, gesture recognition for criminal investigation, video forgery analysis, image tampering detection and many others, in which the identification of specific patterns is required. Deep learning libraries provide many models and approaches for prediction, with the maximum accuracy. 

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